

AD HOC Networks

Basics:

1. What are the various forms of ad-hoc networks?

Mobile ad-hoc networks (MANETs): Self-configuring networks formed by mobile devices without the need for pre-existing infrastructure.

Vehicular ad-hoc networks (VANETs): Networks formed by vehicles equipped with communication capabilities for vehicle-to-vehicle (V2V) or vehicle-to-infrastructure (V2I) communication.

Wireless sensor networks (WSNs): Networks consisting of distributed sensors that communicate wirelessly to monitor physical or environmental conditions.

Mesh networks: Networks where each node can relay data for other nodes, creating a decentralized and self-organizing communication infrastructure.

2. How are ad-hoc networks formed? What are their constituent components?

Ad-hoc networks are formed through self-organization, where nodes autonomously discover and establish connections with neighboring nodes to form a network. Constituent components include mobile devices, sensors, vehicles, or other network-enabled entities equipped with communication capabilities such as wireless transceivers. Nodes communicate directly with nearby nodes or through multi-hop communication if direct communication is not feasible.

3. What characteristics are exhibited by ad-hoc networks?

Characteristics exhibited by ad-hoc networks include:

Dynamic topology: Nodes may join, leave, or move within the network, leading to frequent changes in network topology.

Decentralized control: No centralized infrastructure or control entity; nodes collaborate autonomously to manage network operations.

Limited resources: Nodes often have limited battery power, processing capabilities, and bandwidth, requiring efficient resource utilization.

Multi-hop communication: Communication between nodes may require relaying data through intermediate nodes, especially in large-scale networks or when direct communication is obstructed.

4. What advantages do ad-hoc networks offer?

Flexibility: Ad-hoc networks can be deployed quickly and easily in diverse environments without relying on pre-existing infrastructure.

Robustness: Decentralized control and multi-hop communication enhance resilience to node failures or network disruptions.

Scalability: Ad-hoc networks can scale to accommodate varying numbers of nodes and coverage areas.

Mobility support: Suitable for mobile devices or vehicles where network connectivity is needed while on the move.

Cost-effectiveness: Eliminates the need for expensive infrastructure deployment, making it suitable for temporary or remote deployments.

5. Which applications can benefit from the use of ad hoc networks?

Emergency response and disaster recovery: Providing communication support in disaster-stricken or remote areas where infrastructure may be damaged or unavailable.

Military operations: Facilitating communication between soldiers, vehicles, and command centers in battlefield scenarios.

Sensor networks: Monitoring environmental conditions, industrial processes, or infrastructure health in remote or hazardous locations.

Transportation systems: Enabling vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication for traffic management, collision avoidance, and infotainment services.

6. Discuss the advantages and disadvantages of different routing strategies for ad hoc networks

Proactive routing (e.g., OLSR): Advantages include low latency and quick adaptation to topology changes, but disadvantages include high control overhead and scalability issues in large networks.

Reactive routing (e.g., AODV): Advantages include reduced control overhead and scalability, but disadvantages include higher latency due to route discovery process and potential for stale routes.

Hybrid routing (e.g., ZRP): Combines proactive and reactive approaches to balance trade-offs between control overhead, latency, and scalability.

7. What is the idea of delay-tolerant networking?

The idea of delay-tolerant networking (DTN) is to enable communication in networks where end-to-end connectivity is intermittent or unavailable for extended periods. DTN protocols

store and forward data opportunistically when connectivity is available, relying on store-carry-and-forward mechanisms. This allows communication to occur despite delays or disruptions, making it suitable for scenarios such as remote or rural areas, space communication, or disaster response.

8. What are the advantages and disadvantages of geographical routing.

Advantages of geographical routing in ad-hoc networks include efficient routing based on physical proximity, reduced routing overhead, and scalability. However, disadvantages include the need for location information, susceptibility to localization errors, and challenges in routing over large distances or in dynamic environments with frequent node movements.

MANET Routing

1. What are the challenges of routing in multi-hop ad-hoc networks?

Dynamic topology: Nodes may join, leave, or move within the network, leading to frequent changes in network topology.

Limited resources: Nodes have constrained battery power, processing capabilities, and bandwidth, necessitating efficient routing algorithms.

Multi-hop communication: Routing paths may consist of multiple hops, requiring efficient selection of intermediate nodes for data forwarding.

Interference and fading: Wireless communication is susceptible to interference, signal attenuation, and fading, impacting the reliability of communication links.

Scalability: Routing protocols must scale to accommodate large numbers of nodes and varying network sizes without incurring excessive overhead.

2. What are the differences between link state and distance vector routing protocols?

Differences between link state and distance vector routing protocols:

Link state: Each node maintains a detailed map of the network topology, including information about all nodes and links. Nodes exchange link state advertisements (LSAs) to disseminate topology updates. Shortest path calculations are performed using Dijkstra's algorithm. Examples include OSPF (Open Shortest Path First) and IS-IS (Intermediate System to Intermediate System).

Distance vector: Nodes maintain routing tables containing information about the next hop and distance to reach each destination. Nodes periodically exchange routing updates with neighboring nodes, broadcasting their entire routing table. Shortest path calculations are

performed using the Bellman-Ford algorithm. Examples include RIP (Routing Information Protocol) and BGP (Border Gateway Protocol).

3. What are different classes of ad hoc routing protocols?

Proactive (table-driven) protocols: Maintain up-to-date routing information for all destinations in the network, continuously updating routing tables. Examples include OLSR (Optimized Link State Routing) and DSDV (Destination-Sequenced Distance Vector).

Reactive (on-demand) protocols: Establish routes only when needed, initiating route discovery and maintenance processes dynamically. Examples include AODV (Ad hoc On-Demand Distance Vector) and DSR (Dynamic Source Routing).

Hybrid protocols: Combine proactive and reactive elements to balance overhead and latency. Examples include ZRP (Zone Routing Protocol) and TORA (Temporally Ordered Routing Algorithm).

4. What are differences between reactive and proactive routing protocols?

Differences between reactive and proactive routing protocols:

Reactive: Routes are established on-demand when needed, reducing control overhead but potentially increasing latency due to route discovery process. Examples include AODV and DSR.

Proactive: Routes are continuously maintained and updated, offering low latency but incurring higher control overhead, especially in large networks. Examples include OLSR and DSDV.

5. What is the idea of greedy geographical routing?

The idea of greedy geographical routing is to forward packets towards the destination by choosing the neighbor that is geographically closest to the destination. At each hop, the packet is forwarded to the neighbor that minimizes the distance to the destination based on geographic coordinates. Greedy geographical routing aims to minimize routing overhead and achieve efficient routing paths in ad hoc networks.

6. What types of information are needed to apply geographical routing protocols in ad hoc networks?

To apply geographical routing protocols in ad hoc networks, nodes typically need the following information:

- Geographic coordinates: Each node must know its own geographic location, typically obtained using GPS (Global Positioning System) or other localization techniques.

- Location information of neighboring nodes: Nodes need information about the geographic coordinates of neighboring nodes to determine the next hop towards the destination.
- Destination location: Nodes require the geographic coordinates of the destination node to calculate the direction and distance for forwarding packets.

Reactive Routing – AODV

1. Which routing table entries are established with the RREQ and RREP messages, respectively?

In ad hoc routing protocols like AODV (Ad hoc On-Demand Distance Vector):

- RREQ (Route REQuest) messages establish temporary routing table entries for the reverse path towards the source node.
- RREP (Route REPLY) messages establish permanent routing table entries for the forward path towards the destination node.

2. How is the forwarding and how the reverse path established in the routing tables?

Forwarding and reverse path establishment in routing tables occur as follows:

Forwarding: When a node receives a packet for forwarding, it consults its routing table to determine the next hop towards the destination. The packet is then forwarded to the selected next-hop node.

Reverse path establishment: When a node forwards a packet, it updates its routing table with the reverse path information by recording the source node from which the packet was received as the next hop towards the source.

3. What is the purpose of the sequence number and the hop count maintained in the routing tables?

The purpose of sequence numbers and hop counts in routing tables are:

Sequence number: Helps to uniquely identify and distinguish different versions of routing information, preventing the formation of routing loops and ensuring the freshness of routing updates.

Hop count: Indicates the number of hops or intermediate nodes between the current node and the destination, allowing nodes to select the shortest path towards the destination.

4. What is the purpose sequence number in RREQs and RREPs?

Sequence numbers in RREQs and RREPs serve the purpose of:

RREQs: Sequence numbers in RREQ messages help to ensure the freshness of route requests and prevent the re-discovery of stale routes by specifying the most recent known sequence number for the destination.

RREPs: Sequence numbers in RREP messages help to identify the most recent route to the destination by indicating the sequence number of the route that the replying node is offering.

5. How are multiple RREQs handled by the receiving node? What is the meaning of sequence number and hop count for this handling?

Multiple RREQs received by a node are handled as follows:

Sequence number: The node compares the sequence number in each incoming RREQ with the sequence number of the known route to the destination. If the sequence number in the RREQ is equal to or greater than the sequence number in the routing table, the node updates its routing table with the information from the RREQ.

Hop count: The node considers the hop count in each RREQ to determine the shortest path towards the source. If a RREQ arrives with a shorter hop count than the existing route, the node updates its routing table accordingly.

6. How are link failures detected? What are the mechanisms to repair the link or route?

Link failures are detected through:

Timeout mechanisms: Nodes monitor the activity on their outgoing links and detect failures when expected acknowledgment or activity is not received within a specified timeout period.

Hello messages: Nodes exchange periodic hello messages to verify the connectivity of their neighboring nodes. Failure to receive hello messages from a neighbor indicates a link failure.

Neighboring node notifications: Nodes may receive notifications from neighboring nodes indicating link failures or connectivity issues.

Mechanisms to repair the link or route may include route maintenance processes such as:

Route error propagation: Nodes propagate Route Error (RERR) messages to inform affected nodes about link failures, prompting them to remove or repair stale routes.

Route rediscovery: Nodes initiate route rediscovery processes to establish new routes to affected destinations using alternative paths or nodes.

7. What is the purpose of the RERR message? Which routing tables are modified with this message?

The purpose of the Route Error (RERR) message is to inform neighboring nodes about link failures or unreachable destinations. The routing tables modified with this message include:

The routing table of the node originating the RERR message is updated to remove or invalidate the affected routes.

Neighboring nodes receiving the RERR message update their routing tables to remove or repair routes that are affected by the reported link failures or unreachable destinations.